

ENAUTO Lesson 2 - Python Foundations

Objective

Build an understanding of Python concepts instead of memorizing syntax.

Topics Covered

- Variables
- Data types (str, int, float, bool)
- print()
- Functions
- Parameters
- if / else
- Comparison operators
- Indentation and scope
- Defining vs calling functions
- Debugging syntax and logic errors

Key Concepts

Variables store information.

Functions teach Python a new procedure.

Calling a function executes that procedure.

Python executes code from top to bottom.

Indentation defines which code belongs together.

Example

```
def check_ram(ram):  
    if ram >= 4:  
        print("Enough RAM for lab.")  
    else:  
        print("Not enough RAM.")  
  
check_ram(7.4)  
check_ram(2.0)
```

Big Lessons

Definition vs Execution:

Using def teaches Python a new function. It does not execute it.

Parameters:

A parameter allows one function to work with different values.

Scope:

Indented code belongs to the function. Returning to the left margin ends the function.

Syntax Errors:

Python cannot understand the code.

Logic Errors:

Python understands the code, but the instructions produce the wrong result.

Automation Mindset

Instead of writing one giant script, break the problem into reusable procedures:

```
connect_to_switch()
get_running_config()
save_config()
```

Later, loops will execute those procedures for every device automatically.

Takeaway

You are learning to think like an automation engineer by understanding concepts rather than memorizing syntax. This foundation will make Netmiko, RESTCONF, NETCONF, APIs, and Ansible much easier to learn.